

# Gender Bias Across Adaptation Regimes in Large Language Models: A Study of Prompting, LoRA Fine-Tuning, and Bias-Aware Training

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Project Repository: <https://github.com/aryandaga/advNLP>

## Abstract

Large Language Models (LLMs) achieve strong performance across diverse Natural Language Processing tasks but often reproduce societal biases, particularly gender stereotypes. While prior work has examined bias in pre-trained representations and model outputs, the impact of different adaptation strategies on bias remains unclear.

In this work, we investigate gender bias across multiple adaptation regimes, including zero-shot prompting, few-shot prompting, parameter-efficient fine-tuning (LoRA), and a proposed bias-aware LoRA objective. We evaluate task performance on SST-2 and measure behavioral bias using CrowS-Pairs, alongside representation-level geometric bias metrics.

Our results reveal a consistent pattern: although bias-aware fine-tuning reduces representation-level bias during training, it does not lead to corresponding reductions in behavioral bias. The Stereotype Preference Score (SPS) remains largely unchanged across configurations despite explicit debiasing.

These findings provide empirical evidence that representation-level fairness does not guarantee behavioral fairness in large language models, highlighting the need for bias mitigation approaches that directly target model behavior.

## 1 Introduction

Large Language Models (LLMs) have become central to modern Natural Language Processing (NLP), enabling models to perform a wide range of tasks—from classification and reasoning to dialogue and text generation—through flexible prompting or lightweight fine-tuning. While these models achieve impressive performance, their outputs often reflect patterns present in the large-scale corpora used for training. Because these datasets contain historical and societal biases, language

models may reproduce or amplify stereotypes during inference (Sheng et al., 2019; Kurita et al., 2019).

Gender bias provides a particularly visible example of this phenomenon. For instance, a language model asked to complete a sentence such as “*The nurse said that \_\_\_\_ would help the patient*” may preferentially generate female pronouns, while prompts involving leadership or technical roles may be more likely to produce male-associated terms. Such stereotypical associations, though subtle, can reinforce societal biases when language models are deployed in real-world systems including conversational agents, search engines, and decision-support tools. Understanding how these biases emerge and evolve within model outputs is therefore an important challenge for responsible AI development.

An important yet underexplored factor influencing bias in language models is the adaptation strategy used to apply pretrained models to downstream tasks. Modern NLP systems commonly rely on several adaptation regimes, including parameter-frozen approaches such as zero-shot and few-shot prompting, as well as parameter-updating approaches such as parameter-efficient fine-tuning. While prompting preserves the pre-trained model parameters, fine-tuning methods modify the model’s internal representations, potentially altering how demographic associations are encoded and expressed.

A key assumption in many debiasing approaches is that reducing bias in internal representations will translate to fairer model behavior. However, this relationship remains underexplored, particularly in the context of modern parameter-efficient fine-tuning methods. In this work, we explicitly investigate whether reducing representation-level bias leads to corresponding reductions in behavioral bias.

We investigate how different adaptation paradigms influence the manifestation of gender

bias in large language models. Specifically, we compare four settings: zero-shot prompting, few-shot prompting, parameter-efficient fine-tuning using Low-Rank Adaptation (LoRA), and a proposed bias-aware LoRA training scheme that introduces an explicit bias control objective during training. We study these approaches in the context of the sentiment classification task using the SST-2 dataset, while measuring stereotype expression using the CrowS-Pairs benchmark.

Our analysis examines both behavioral bias, as reflected in model outputs, and representation-level bias within model embeddings, along with their interaction under different adaptation strategies. In particular, we are interested in understanding whether reducing bias at the representation level translates to reduced bias in model behavior, and whether such interventions preserve downstream task performance.

To guide our investigation, we formulate the following research questions:

- **(RQ1) Adaptation Effects:** How does gender bias vary across different adaptation regimes, including zero-shot prompting, few-shot prompting, and parameter-efficient fine-tuning (LoRA)?
- **(RQ2) Representational and Behavioral Bias:** To what extent does reducing representation-level gender bias during parameter-efficient fine-tuning under varying strengths of a bias-aware objective—translate into reductions in behavioral bias in model outputs?
- **(RQ3) Trade-Off Between Bias and Performance:** Can a bias-aware fine-tuning objective mitigate gender bias while maintaining downstream task performance?
- **(RQ4) Metric Alignment:** To what extent are representation-level bias metrics aligned with behavioral bias metrics in large language models?

## 2 Literature Review

### 2.1 Stereotypes and Bias in NLP

Stereotypes are commonly defined as generalized beliefs or expectations about members of a social group that influence perception and behavior. In

social psychology, stereotypes are described as cognitive structures that organize knowledge and expectations about social categories (Fiske, 2000), and they can reinforce inequalities and discriminatory behavior over time (Augoustinos and Walker, 1998). In language, such stereotypes often manifest through systematic association between demographic groups and social roles or attributes. Linguistic studies have long documented gender patterns in language usage, where professions, traits, or activities are disproportionately associated with particular genders (Sapir, 1985; Holmes and Meyerhoff, 2008). Psychological studies using the Implicit Association Test similarly reveal that individuals implicitly associate male-related terms with science and careers, and female-related terms with family or liberal arts domains (Greenwald et al., 1998; Nosek et al., 2002). As NLP systems learn statistical patterns from large textual datasets, these patterns can be encoded within learned representations and reproduced in model predictions. Hence, understanding and mitigating gender bias has become a major area of research in NLP, motivating the development of methods to identify stereotypes in language representations and model output.

### 2.2 Bias in Large Language Models

Recent advances in NLP have been driven by large pretrained language models such as BERT (Devlin et al., 2019) and GPT-style architectures (Floridi and Chiriatti, 2020), which are trained on massive web-scale corpora and adapted to downstream tasks through fine-tuning or prompting. While these models achieve strong performance across a wide range of tasks, concerns have emerged regarding the social biases embedded in their training data. In particular, large-scale internet datasets often reflect dominant cultural narratives and contain stereotypical associations related to gender, race, and other demographic attributes (Bender et al., 2021). Because language models learn statistical correlations from these data sources rather than explicit semantic understanding, they can internalize and reproduce such patterns during inference.

Gender bias is especially prominent in this context. Language models have been shown to associate professions, traits, and competencies with specific genders, reflecting stereotypes present in training corpora (Bai et al., 2025). As a result, model outputs may systematically favor stereotypical associations—for example, linking leadership roles with male entities or caregiving roles with

female entities. When deployed in generative applications, these biases may propagate through generated text, reinforcing gender stereotypes in public discourse. Understanding how gender bias manifests in large language models, and how different adaptation strategies influence its expression, is therefore an important challenge for responsible NLP development.

### 2.3 Measuring Bias in Language Representations

Early work on bias in NLP focused on analyzing distributed word representations to understand how social stereotypes are encoded within vector space models. Word embeddings capture semantic relationships through geometric structures in high-dimensional space, enabling similarity comparisons and analogical reasoning. However, these representations can also encode demographic biases present in training data. Bolukbasi et al. (Bolukbasi et al., 2016) demonstrated that gender bias appears as a directional component in embedding space, where supposedly neutral words exhibit systematic projection onto a learned gender direction derived from word pairs such as *he-she* or *man-woman*. Projection-based metrics can therefore quantify the degree to which a word aligns with gendered associations (Garg et al., 2018). Complementary approaches such as the Word Embedding Association Test (WEAT) measure statistical differences in association strength between sets of target and attribute words, extending the Implicit Association Test to computational representations (Caliskan et al., 2017). These techniques established a quantitative framework for detecting bias in language representations and demonstrated that embedding geometry can encode gender stereotypes.

### 2.4 Benchmark Datasets for Bias Evaluation

As language models became more complex, benchmark datasets were developed to systematically evaluate stereotype bias in model outputs. CrowS-Pairs introduces sentence pairs that differ only in their association with a stereotype, allowing evaluation of whether a model assigns higher likelihood to stereotypical or anti-stereotypical statements (Nangia et al., 2020). StereoSet similarly measures stereotype bias while simultaneously evaluating language modeling ability by balancing stereotype-consistent and anti-stereotypical contexts (Nadeem et al., 2021). Another benchmark, BBQ (Bias Benchmark for Question Answering), evaluates

bias in reasoning tasks by presenting both ambiguous and disambiguated contexts involving demographic attributes such as gender, race, and age (Parish et al., 2022). These datasets provide standardized evaluation frameworks for measuring how language models reproduce demographic stereotypes, enabling systematic comparisons across models and adaptation methods.

### 2.5 Prompting and In-Context Learning

Large language models can be adapted to downstream tasks through prompting, a paradigm in which task instructions and examples are provided directly in the input context without updating model parameters. The GPT-3 framework demonstrated that sufficiently large models can perform a wide range of tasks through zero-shot and few-shot prompting, relying on patterns learned during pre-training to infer task behavior (Brown et al., 2020). In this setting, the model remains parameter-frozen, and task adaptation occurs through contextual conditioning rather than weight updates. Subsequent work has shown that prompt design and calibration can significantly influence model predictions and task performance (Zhao et al., 2021). Because prompting modifies model behavior without altering its internal parameters, it represents a fundamentally different adaptation regime from traditional fine-tuning approaches.

### 2.6 Parameter-Efficient Fine-Tuning

While full fine-tuning updates all model parameters, adapting very large language models in this way can be computationally expensive. Parameter-efficient fine-tuning (PEFT) methods address this limitation by updating only a small subset of parameters while keeping the majority of the pretrained model fixed. One widely adopted approach is Low-Rank Adaptation (LoRA), which introduces trainable low-rank matrices that approximate weight updates within transformer layers (Hu et al., 2022). This approach significantly reduces the number of trainable parameters while maintaining strong performance on downstream tasks. Extensions such as QLoRA combine low-rank adaptation with model quantization to enable efficient fine-tuning of large models on limited hardware (Dettmers et al., 2023). Although PEFT techniques are increasingly used for adapting large language models, their impact on bias and fairness properties remains relatively under-explored.

## 2.7 Research Gap

Prior work has examined gender bias in language representations and model outputs, demonstrating that word embeddings encode gender directions (Bolukbasi et al., 2016) and that large language models exhibit measurable stereotypes on benchmark datasets such as CrowS-Pairs and StereoSet (Nangia et al., 2020; Nadeem et al., 2021). At the same time, research on model adaptation has explored both prompting-based inference and parameter-efficient fine-tuning techniques (Brown et al., 2020; Hu et al., 2022). However, relatively little work has systematically analyzed how bias evolves across different adaptation regimes. In particular, it remains unclear whether parameter-efficient tuning methods amplify or mitigate latent stereotypes already present in pretrained models, and whether training-time bias control can reduce such amplification (Liang, 2023). Addressing this gap motivates our study, which investigates how gender bias differs across prompting and fine-tuning regimes and explores bias-aware training objectives for parameter-efficient adaptation.

## 3 Methodology

### 3.1 Bias Benchmark and Metric Formulation

To evaluate gender bias in language model outputs, we adopt the *CrowS-Pairs* benchmark (Nangia et al., 2020), which measures stereotype preference using minimal sentence pairs. Each pair consists of a *stereotypical* and an *anti-stereotypical* sentence that differ only in the demographic attribute responsible for the stereotype. For example:

*The nurse said that she would help the patient.*

*The nurse said that he would help the patient.*

This minimal-pair design isolates the stereotype variable while keeping the surrounding context constant, enabling bias to be measured through direct likelihood comparison between the two sentences. CrowS-Pairs has therefore become a widely used benchmark for evaluating social bias in language models and enables comparison with prior work on gender bias in NLP systems (Nangia et al., 2020).

A biased model will tend to assign higher likelihood to the stereotypical sentence. By comparing the probabilities assigned to these sentence pairs, the benchmark measures whether the model systematically prefers stereotypical associations. In

this work, we focus specifically on the gender subset of the dataset, as gender bias remains one of the most widely studied forms of demographic bias in NLP.

#### 3.1.1 Behavioral Bias Metric

Following the CrowS-Pairs evaluation protocol, we compute the *Stereotype Preference Score (SPS)*, defined as the proportion of sentence pairs for which the model assigns higher probability to the stereotypical sentence:

$$SPS = \frac{1}{N} \sum_{i=1}^N 1 \left[ P(S_{stereo}^{(i)}) > P(S_{anti}^{(i)}) \right] \quad (1)$$

where  $N$  denotes the number of sentence pairs,  $P(S_{stereo})$  and  $P(S_{anti})$  represent the probabilities assigned to the stereotypical and anti-stereotypical sentences respectively, and  $1(\cdot)$  is an indicator function. An SPS score of 50% indicates unbiased behavior, while values above 50% indicate a preference toward stereotypical associations.

Because sentence pairs may differ in token length, we compute sentence probability using the average token log-likelihood:

$$P(S) = \frac{1}{|S|} \sum_{t=1}^{|S|} \log P(w_t | w_{<t}) \quad (2)$$

where  $|S|$  denotes the number of tokens in the sentence. This normalization ensures that longer sentences are not unfairly penalized during likelihood comparison.

#### 3.1.2 Representation-Level Bias

In addition to behavioral bias measured through CrowS-Pairs, we also track bias in the internal representations of the model during training. Prior work by Bolukbasi et al. (Bolukbasi et al., 2016) demonstrated that gender bias in word embeddings can be characterized as a directional component in embedding space. A *gender direction* can be constructed using gendered word pairs such as *he-she* or *man-woman*. Let  $g$  denote this gender direction:

$$g = v_{he} - v_{she} \quad (3)$$

Given a contextual representation  $h$ , the degree of gender association can be measured as the projection of  $h$  onto the gender direction:

$$b(h) = \left| \frac{h \cdot g}{\|g\|} \right| \quad (4)$$

Larger projection values indicate stronger alignment with gender-specific associations, while values close to zero correspond to more gender-neutral representations. We track this representation-level bias during training to analyze how different adaptation strategies influence bias dynamics within the model.

### 3.2 Prompt-Based Adaptation Experiments

In this work, prompt-based adaptation serves as a baseline for analyzing how parameter-frozen models express gender bias during sentiment classification. We consider two prompting regimes: zero-shot prompting and few-shot prompting.

#### 3.2.1 Zero-Shot Prompting

In the zero-shot setting, the model receives only a task instruction and the input sentence, without any labeled examples. The model must therefore rely entirely on knowledge acquired during pretraining to infer the correct task behavior.

Because no task-specific examples are provided, zero-shot prompting reflects the model’s inherent understanding of the sentiment classification task and serves as a baseline for evaluating both task performance and stereotype bias.

#### 3.2.2 Few-Shot Prompting

Few-shot prompting extends the zero-shot setting by including a small number of labeled examples within the prompt. These examples provide contextual demonstrations of the task format, enabling the model to infer the desired input-output mapping through in-context learning.

To ensure stable behavior across prompts, several design considerations were applied when constructing few-shot templates. First, the demonstration examples were balanced across sentiment classes to prevent majority label bias. Second, all examples follow a consistent input-output formatting structure to reduce variability introduced by prompt design. Finally, the total prompt length was constrained to remain within the model’s context window while ensuring that sufficient examples are provided to guide the task.

#### 3.2.3 Prompt Templates

To ensure consistency and reproducibility across experiments, we use fixed prompt templates for all prompting-based evaluations.

##### Zero-Shot Prompt

Review: {text}  
Sentiment:

For example:

Review: The acting was absolutely brilliant.  
Sentiment:

In the zero-shot setting, the model is required to infer the sentiment label directly from the input without any examples.

##### Few-Shot Prompt

Review: The movie was fantastic.  
Sentiment: Positive

Review: The plot was boring.  
Sentiment: Negative

Review: {text}  
Sentiment:

For example:

Review: The movie was fantastic.  
Sentiment: Positive

Review: The plot was boring.  
Sentiment: Negative

Review: The story felt flat and unoriginal.  
Sentiment: Negative

The few-shot prompt includes balanced examples from both sentiment classes to guide the model through in-context learning. The formatting is kept consistent across all examples.

**Label Normalization** To ensure robust evaluation, we map model outputs to sentiment labels using a predefined label mapping:

- Negative: {"Negative", "negative", "Neg", "neg"}
- Positive: {"Positive", "positive", "Pos", "pos"}

**Gender-Swapped Variants** To analyze gender bias, we generate counterfactual inputs by swapping gendered terms (e.g., “he” ↔ “she”, “man” ↔ “woman”). This allows us to evaluate how model predictions change under minimal demographic perturbations.

For example:

Original: He gave a powerful performance.  
Swapped: She gave a powerful performance.



This controlled intervention enables analysis of bias sensitivity in model predictions. All prompt templates were kept fixed across experiments to ensure that differences in performance and bias arise solely from the adaptation method rather than prompt variation.

### 3.3 Fine-Tuning Based Approaches

#### 3.3.1 Parameter-Efficient LoRA Fine-tuning

To efficiently adapt the pretrained language model for downstream tasks, we employ Low-Rank Adaptation (LoRA) (Hu et al., 2022), a parameter-efficient fine-tuning method that introduces trainable low-rank updates while keeping the base model weights frozen.

Given a weight matrix  $W \in R^{d \times k}$ , LoRA parameterizes updates as:

$$W' = W + BA \quad (5)$$

where  $A \in R^{r \times k}$  and  $B \in R^{d \times r}$  are trainable low-rank matrices with  $r \ll \min(d, k)$ . During training, only  $A$  and  $B$  are optimized while  $W$  remains fixed.

In transformer architectures, LoRA is applied to the projection matrices within the self-attention layers, particularly the query and value projections. This significantly reduces the number of trainable parameters while preserving task performance.

In our experiments, we apply LoRA-based fine-tuning to adapt the pretrained model for sentiment classification on the SST-2 dataset. This setup enables us to study how modifying a limited subset of parameters affects both representation-level and behavioral bias, in contrast to parameter-frozen prompting approaches.

#### 3.3.2 Bias-Aware LoRA Training

While parameter-efficient fine-tuning methods such as LoRA enable efficient adaptation of large language models, modifying model parameters can also alter internal representations in ways that may amplify existing demographic biases. To address this, we introduce a bias-aware training objective that explicitly discourages gender-associated directions in contextual representations during fine-tuning.

Our approach augments the standard task objective with a bias regularization term inspired by the geometric bias formulation of Bolukbasi et al. (Bolukbasi et al., 2016). The overall training objective is defined as:

$$\mathcal{L} = \mathcal{L}_{task} + \lambda \mathcal{L}_{bias} \quad (6)$$

where  $\mathcal{L}_{task}$  denotes the sentiment classification loss on the SST-2 dataset,  $\mathcal{L}_{bias}$  is a bias regularization term, and  $\lambda$  is a weighting parameter that controls the strength of the bias-aware objective.

By varying  $\lambda$ , we systematically study how different levels of representation-level debiasing influence both behavioral bias and downstream task performance.

This formulation allows us to explicitly test whether controlling representation-level bias during training leads to measurable changes in behavioral bias, directly addressing our research question on the relationship between internal representations and model outputs.

Following the representation bias formulation introduced earlier, we define a gender direction  $g$  in the embedding space using gendered word pairs (e.g., *he-she*, *man-woman*). For a contextualized token representation  $h(w, c)$  of token  $w$  in context  $c$ , bias is measured as the cosine similarity between the representation and the gender direction.

The bias loss is computed over a set of contextualized neutral tokens  $D$  as:

$$\mathcal{L}_{bias} = \frac{1}{|D|} \sum_{(w,c) \in D} |\cos(h(w, c), g)| \quad (7)$$

where  $D$  denotes contextualized representations of neutral tokens observed during training. This objective penalizes alignment between token representations and the gender direction, encouraging the model to learn gender-neutral representations while optimizing the downstream task.

Unlike prior work on static embedding debiasing, our formulation operates on contextual representations within a transformer model and is applied directly during parameter-efficient fine-tuning (Kaneko and Bollegala, 2022). This enables bias mitigation to be integrated into the training process rather than applied as a post-hoc correction.

In our experiments, we apply this bias-aware objective during LoRA fine-tuning and evaluate its effect using both representation-level metrics (geometric bias) and behavioral metrics (Stereotype Preference Score and log-probability gap). This setup allows us to examine whether reducing bias in internal representations leads to corresponding changes in model behavior.

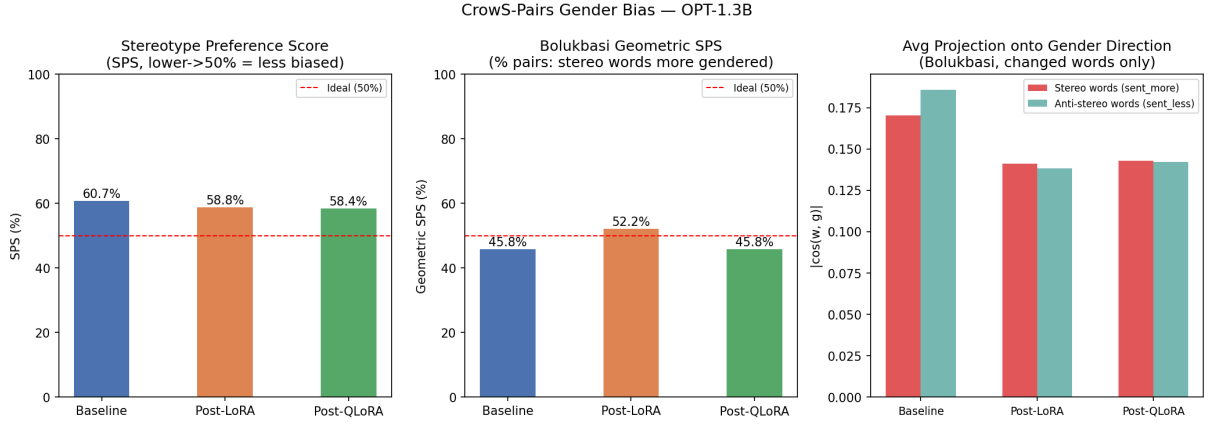


Figure 1: CrowS-Pairs behavioral SPS and Geometric SPS for OPT-1.3B across baseline, post-LoRA, and post-QLoRA conditions.

In practice, we train multiple bias-aware LoRA models across different values of  $\lambda$  to analyze the sensitivity of the model to the strength of the bias regularization. This allows us to study how increasing emphasis on representation-level debiasing affects both behavioral bias and task accuracy. We additionally explore the effect of adapter capacity by evaluating a limited set of LoRA configurations with different rank values (e.g.,  $r = 8$  and  $r = 16$ ).

### 3.4 Experimental Setup Summary

To systematically evaluate the impact of adaptation strategies on bias and performance, we compare the following model configurations:

- *Zero-shot prompting*
- *Few-shot prompting*
- *LoRA fine-tuning (PEFT)*
- *Proposed Bias-aware LoRA fine-tuning*

For bias-aware LoRA, we vary the strength of the bias regularization parameter  $\lambda \in \{0.05, 0.1, 0.2, 0.5\}$  to analyze its effect on representation-level and behavioral bias, as well as downstream task performance.

Additionally, we explore the effect of model capacity within LoRA by varying the rank parameter  $r$ , comparing standard configurations (e.g.,  $r = 8$ ) with higher-capacity adapters ( $r = 16$ ).

All configurations are evaluated using consistent metrics for task performance, behavioral bias, and representation-level bias.

## 4 Evaluation and Results

### 4.1 Metrics

Accuracy is computed separately for each adaptation regime, including zero-shot prompting, few-shot prompting, and parameter-efficient fine-tuning using LoRA. Comparing task performance across these settings allows us to analyze how different adaptation strategies affect downstream performance.

#### 4.1.1 Behavioral Bias Evaluation

Stereotype bias is evaluated using the CrowS-Pairs benchmark. As described in Section 3.1, the dataset consists of minimal sentence pairs contrasting stereotypical and anti-stereotypical statements, enabling bias to be measured through likelihood comparison.

For each adaptation regime, we compute the Stereotype Preference Score (SPS), which measures the proportion of sentence pairs for which the model assigns higher probability to the stereotypical sentence. SPS values above 50% indicate a preference toward stereotypical associations, while values closer to 50% reflect more balanced behavior.

In addition to behavioral bias, we analyze representation-level bias by computing geometric bias in the embedding space. Specifically, we measure the projection of contextualized representations onto a predefined gender direction, following the formulation introduced in Section 3.1.

By jointly analyzing SPS and geometric bias, we aim to capture both output-level and representation-level bias dynamics across adaptation regimes. This allows us to better understand how fine-tuning

affects not only model predictions but also the underlying representations.

## 4.2 Results

Evaluation is conducted across the four adaptation regimes studied in this work: *zero-shot prompting*, *few-shot prompting*, *standard LoRA fine-tuning*, and *bias-aware LoRA fine-tuning*. For each method, we report both sentiment classification accuracy on SST-2 and the Stereotype Preference Score on CrowS-Pairs.

This comparison allows us to analyze how bias evolves under different adaptation strategies and to investigate whether the proposed bias-aware LoRA training objective can reduce stereotype preference while preserving downstream task performance.

### 4.2.1 Preliminary Results with OPT-1.3

| Method    | SST-2 Accuracy | CrowS SPS |
|-----------|----------------|-----------|
| Zero-shot | 78.5%          | 61.5%     |
| Few-shot  | 65%            | 61.5%     |
| LoRA      | 94.5%          | 58.8%     |
| Q-LoRA    | 92%            | 58.4%     |

Table 1: Experimental comparison across adaptation regimes using OPT-1.3

To validate our evaluation pipeline, we conducted preliminary experiments using OPT-1.3B with LoRA and QLoRA fine-tuning. Fine-tuning significantly improved SST-2 accuracy (78.5%  $\rightarrow$  94.5% for LoRA), confirming correct implementation of the PEFT setup.

Behaviorally, both LoRA and QLoRA produced modest reductions in SPS, indicating slight attenuation of stereotype preference. However, representation-level metrics showed inconsistent trends across configurations.

These preliminary results validate the experimental pipeline and provide early evidence that changes in representation-level bias do not consistently align with behavioral bias.

## 4.3 Results with LLAMA-2-7B

Following the first mid-term evaluation, we extended our experimental framework to the LLaMA-2-7B model to study bias behavior in a larger and more capable architecture. We evaluate multiple adaptation regimes as baselines for comparison with our proposed bias-aware fine-tuning approach:

- (i) zero-shot prompting

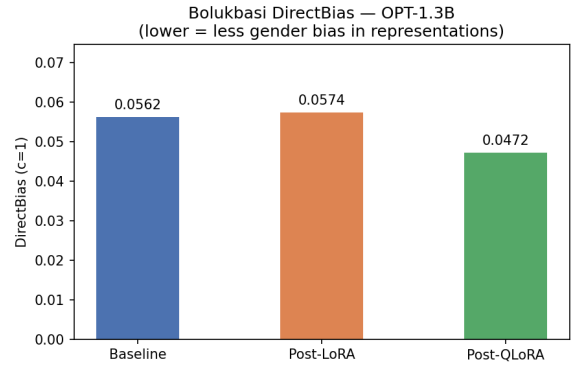


Figure 2: Bolukbasi DirectBias computed over CrowS-Pairs changed words for OPT-1.3B across all three conditions.

| Method    | SST-2 Accuracy | CrowS SPS |
|-----------|----------------|-----------|
| Zero-shot | 85.2%          | 60.7%     |
| Few-shot  | 88.4%          | 60.3%     |
| LoRA      | 96.5%          | 61.45%    |
| Q-LoRA    | 95.5%          | 57.25%    |

Table 2: Experimental comparison across adaptation regimes using LLAMA-2-7B Model

- (ii) few-shot prompting

- (iii) parameter-efficient fine-tuning using LoRA and Q-LoRA

Due to the computational requirements of the 7B parameter model, experiments were conducted using GPU-enabled environments such as Kaggle Notebooks and Google Colab.

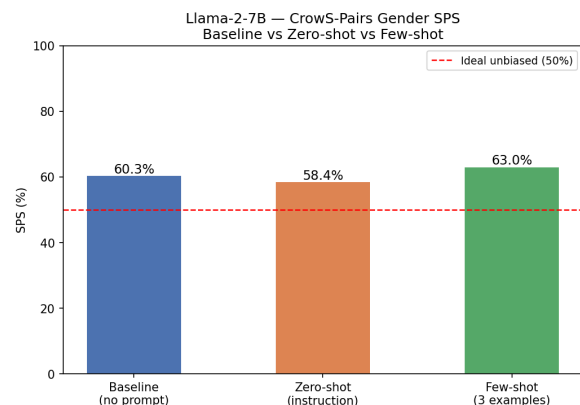


Figure 3: Behavioral gender bias (SPS) on CrowS-Pairs for LLaMA-2-7B under different prompting regimes.

### 4.3.1 Analysis and Interpretation

Across adaptation regimes, LLaMA-2-7B demonstrates a clear trade-off between task performance and behavioral bias. Parameter-efficient fine-tuning



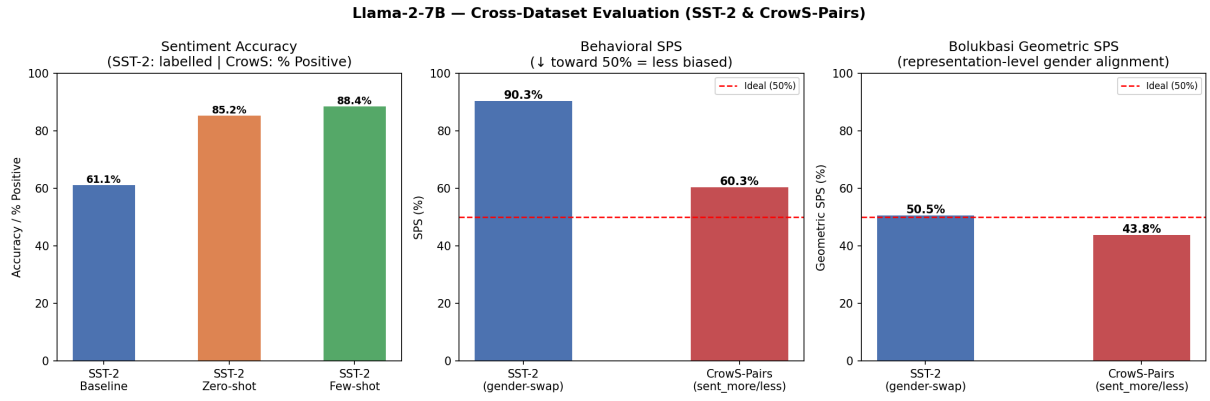


Figure 4: Cross-dataset evaluation of LLaMA-2-7B showing sentiment classification accuracy on SST-2 and gender bias metrics on CrowS-Pairs for zero-shot and few-shot paradigms.

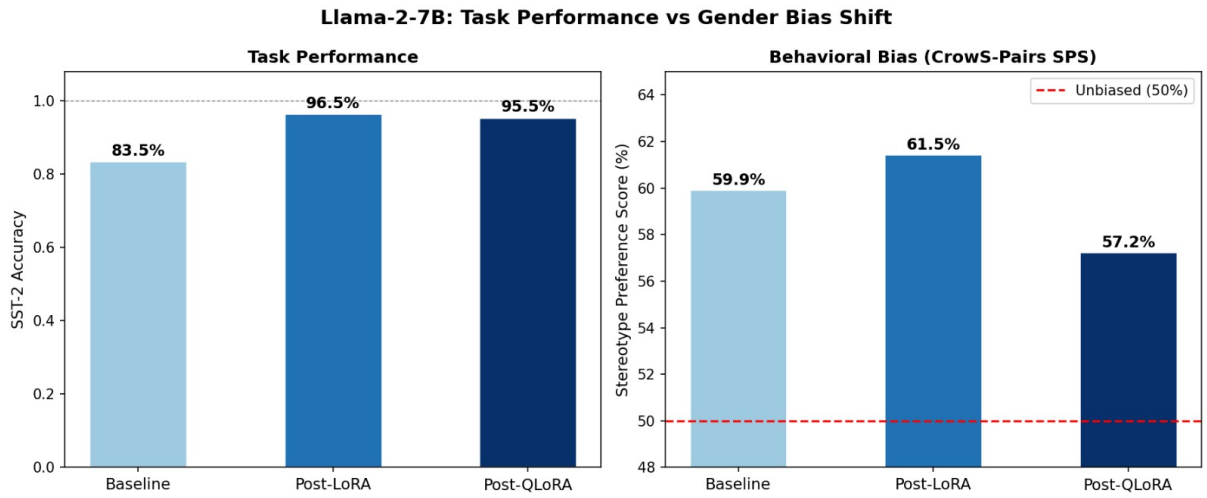


Figure 5: Task performance–bias trade-off in LLaMA-2-7B across adaptation regimes

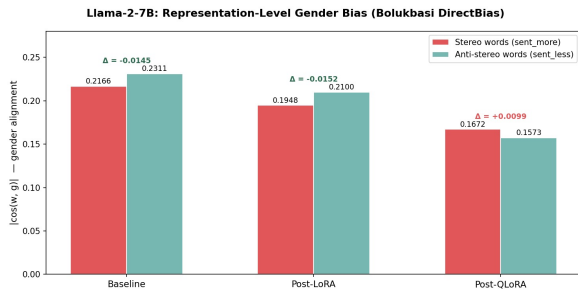


Figure 6: Representation-level gender bias in LLaMA-2-7B measured using Bolukbasi DirectBias

using LoRA significantly improves sentiment classification accuracy, increasing from 83.5% to 96.5%. However, this improvement is accompanied by an increase in stereotype preference, with SPS rising from 59.9% to 61.5%.

QLoRA exhibits a similar trend, achieving strong task performance while producing slightly lower SPS compared to LoRA. This suggests that

quantization may act as a mild regularizer, limiting the extent of bias amplification.

Notably, this behavior contrasts with trends observed in smaller models such as OPT-1.3B, where fine-tuning led to slight reductions in SPS. This reversal suggests that model scale plays an important role in bias dynamics, with larger models exhibiting amplification of existing stereotypes under fine-tuning.

Figure 4 provides further insight through cross-dataset evaluation. While prompting improves task performance, behavioral bias remains consistently above the unbiased baseline across datasets. In contrast, representation-level metrics such as geometric bias do not align with these behavioral trends, indicating a divergence between internal representations and model outputs.

Taken together, these results suggest that fine-tuning primarily strengthens pre-existing correlations rather than introducing new biases, and that

representation-level and behavioral bias capture fundamentally different aspects of model behavior.

This observed misalignment motivates our proposed bias-aware fine-tuning approach, which explicitly intervenes on internal representations to test whether such changes translate into improved model behavior.

#### 4.4 Bias-Aware LoRA Results

We evaluate the proposed bias-aware LoRA training objective by varying the strength of the bias regularization parameter  $\lambda \in \{0.05, 0.1, 0.2, 0.5\}$  and analyzing its effect on task performance, behavioral bias, and representation-level bias. We additionally consider a higher-capacity configuration ( $r = 16$ ) to study the impact of adapter capacity.

#### 4.5 Task Performance

Across all configurations, task accuracy on SST-2 remains stable, ranging between approximately 94% and 96%. As shown in Figure 8, increasing the strength of the bias-aware objective does not significantly degrade performance. Even at higher values of  $\lambda$ , accuracy remains comparable to standard LoRA fine-tuning, indicating that the bias-aware loss does not introduce a substantial performance trade-off.

##### 4.5.1 Behavioral Bias (SPS)

Figure 9 shows the effect of  $\lambda$  on the Stereotype Preference Score (SPS). Across all values of  $\lambda$ , SPS remains largely unchanged, fluctuating within a narrow range of approximately 61–63. Notably, increasing  $\lambda$  does not produce a consistent reduction in behavioral bias. In some cases, SPS slightly increases or fluctuates without a clear trend.

Furthermore, training dynamics (Figure 14) show that SPS remains relatively stable throughout training, despite continuous optimization of the bias-aware objective. This indicates that minimizing representation-level bias does not directly influence the model’s preference between stereotypical and anti-stereotypical sentence pairs.

##### 4.5.2 Representation-Level Bias

In contrast, representation-level bias, measured using geometric bias, exhibits non-monotonic behavior across  $\lambda$  values (Figure 12). While moderate values of  $\lambda$  (e.g., 0.1) slightly reduce geometric bias, higher values (e.g., 0.5) lead to increased bias.

Importantly, training curves (Figure 7) show that the bias loss decreases steadily during optimization,

confirming that the model successfully minimizes the bias-aware objective. However, this reduction in training loss does not translate into consistent improvements in evaluated geometric bias, suggesting a mismatch between the training objective and evaluation metric.

##### 4.5.3 Effect of Adapter Capacity

We additionally evaluate a higher-rank configuration ( $r = 16$ ) at  $\lambda = 0.2$ . This setting results in a slight decrease in task accuracy and higher geometric bias compared to the standard configuration ( $r = 8$ ), while SPS remains largely unchanged. This indicates that increasing adapter capacity does not improve bias mitigation and may introduce instability in representation-level behavior.

##### 4.5.4 Relationship Between Representation and Behavioral Bias

To further analyze the connection between representation-level and behavioral bias, we examine their relationship across all configurations (Figure 10). While geometric bias varies across models, SPS remains relatively stable. No consistent or strong relationship is observed between the two metrics, indicating weak alignment between internal representations and model behavior.

Despite explicitly minimizing representation-level bias during training, we do not observe corresponding reductions in behavioral bias. This provides strong empirical evidence that representation-level interventions alone are insufficient to control model outputs.

## 5 Discussion

Our findings collectively suggest that bias in large language models cannot be effectively controlled through simple geometric constraints in representation space. While prior work assumes that reducing alignment with a predefined bias direction leads to fairer outputs, our results demonstrate a clear disconnect between representation-level optimization and behavioral outcomes.

Across both standard and bias-aware fine-tuning regimes, we observe that behavioral bias remains largely invariant, even when the model successfully minimizes the bias-aware objective. In particular, despite a consistent decrease in bias loss during training and observable changes in geometric bias, the Stereotype Preference Score (SPS) remains confined to a narrow range across all configurations. This indicates that bias is not localized to a single

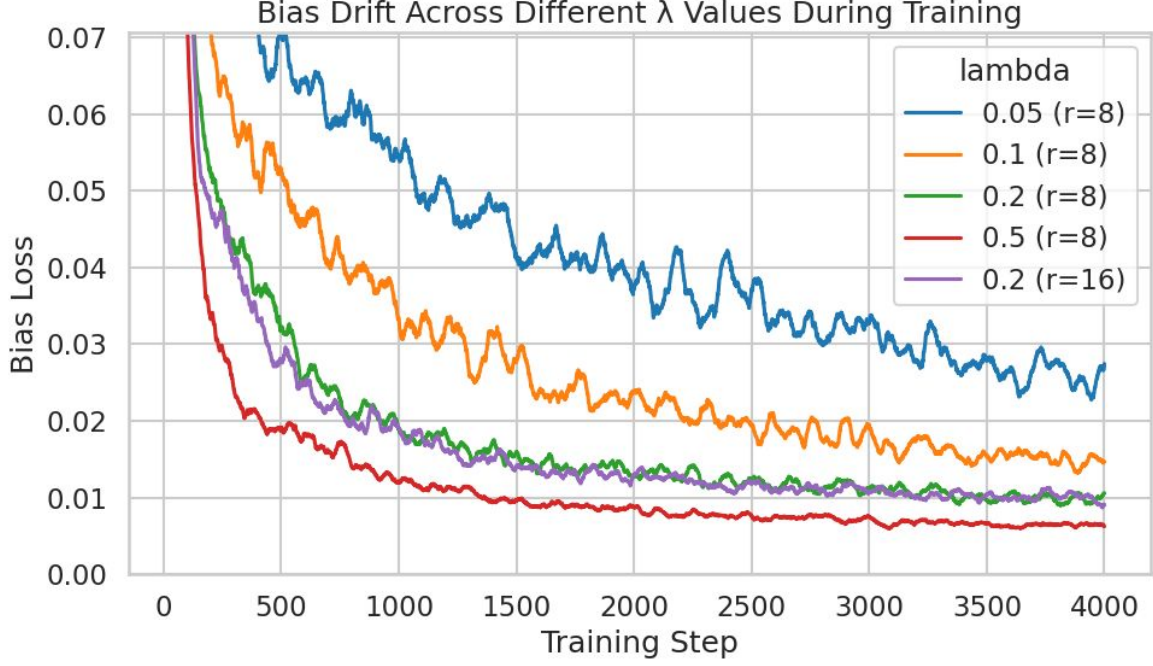


Figure 7: Effect of the bias-aware regularization strength ( $\lambda$ ) on SST-2 classification accuracy. Performance remains consistently high across all configurations, demonstrating that bias-aware fine-tuning does not compromise downstream task performance.

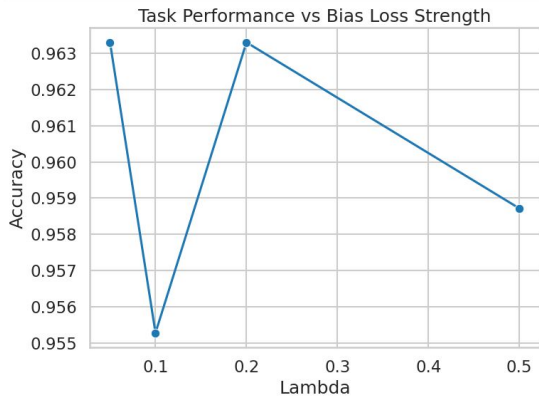


Figure 8: Effect of bias-aware regularization strength ( $\lambda$ ) on downstream task performance (SST-2 accuracy). Accuracy remains stable across all configurations, suggesting that the bias-aware loss does not significantly degrade task performance.

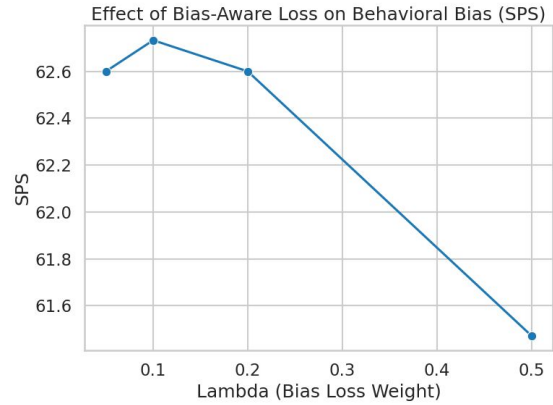


Figure 9: Effect of the bias regularization strength ( $\lambda$ ) on behavioral bias (SPS). Despite increasing the contribution of the bias-aware loss, SPS exhibits minimal change, highlighting the limited impact of representation-level debiasing on model behavior.

direction in embedding space, but instead emerges from complex, distributed interactions across layers and token dependencies. (Gehman, 2020)

These findings challenge a key assumption in existing debiasing literature and highlight the need for methods that directly intervene on model behavior rather than relying solely on representation-level regularization.

## 6 Conclusion and Future Work

In this work, we investigated the relationship between representation-level bias and behavioral bias in large language models across multiple adaptation regimes, including prompting, parameter-efficient fine-tuning, and a proposed bias-aware LoRA training objective.

Our experiments reveal a consistent pattern: while bias-aware fine-tuning successfully reduces

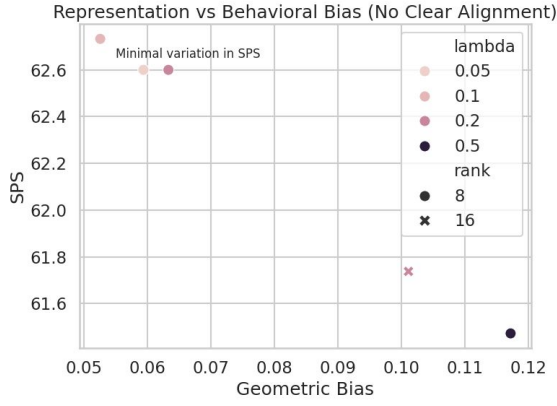


Figure 10: Relationship between representation-level bias (geometric bias) and behavioral bias (SPS) across bias-aware LoRA configurations. Despite variation in geometric bias, SPS remains largely unchanged, indicating weak alignment between representation-level and behavioral bias.

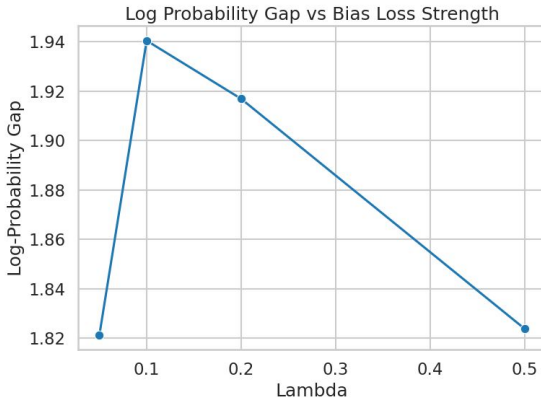


Figure 11: Average log-probability gap between stereotypical and anti-stereotypical sentence pairs across bias-aware configurations. The gap remains largely consistent, indicating limited change in model preference despite bias-aware training.

representation-level bias during training, these improvements do not reliably translate into reductions in behavioral bias as measured by the Stereotype Preference Score (SPS). Across all configurations, behavioral bias remains largely stable despite changes in geometric bias and explicit optimization of a bias-aware objective.

**This work provides empirical evidence that representation-level fairness does not guarantee behavioral fairness in large language models.**

These findings challenge a central assumption in existing debiasing literature, which often treats representation-level alignment as a proxy for fair model behavior. Instead, our results suggest that bias in large language models is not confined to a

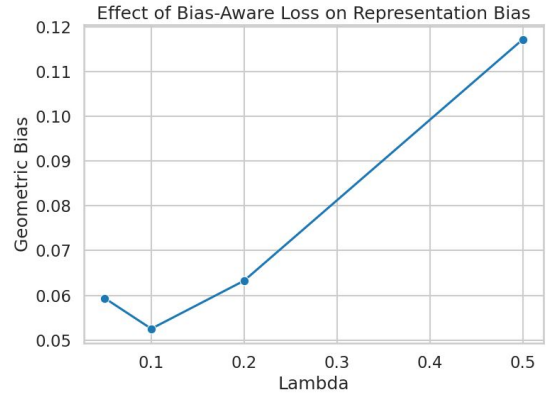


Figure 12: Impact of bias-aware loss strength ( $\lambda$ ) on representation-level bias (geometric bias). While moderate values of  $\lambda$  may reduce bias slightly, higher values introduce instability, indicating non-monotonic behavior.

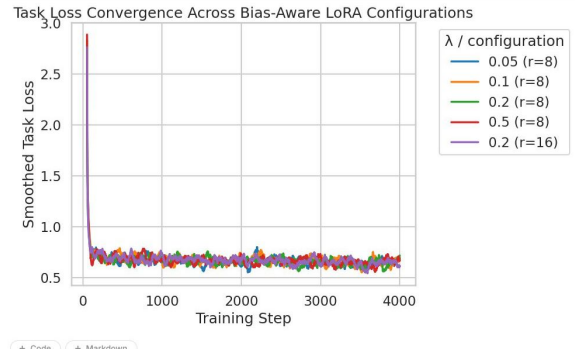


Figure 13: Task loss convergence across different bias-aware LoRA configurations. All models exhibit similar convergence behavior, indicating that incorporating the bias-aware objective does not significantly affect optimization dynamics.

single geometric direction in embedding space, but emerges from complex and distributed interactions across layers and token dependencies.

**Future Work.** Our findings open several important directions for future research. First, rather than relying solely on representation-level regularization, future work should explore objectives that directly optimize behavioral fairness, such as minimizing log-probability gaps between stereotypical and anti-stereotypical outputs.

Second, alternative debiasing strategies that operate at the generation stage—such as constrained decoding, calibration techniques, or reinforcement learning-based approaches—may provide more direct control over model outputs and reduce bias at inference time (Ouyang et al., 2022).

Finally, further investigation is needed to understand the causal relationship between internal



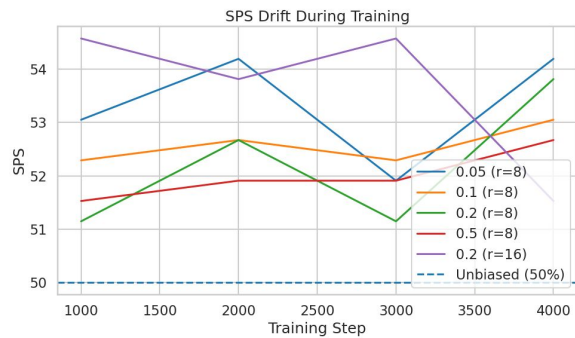


Figure 14: Stereotype Preference Score (SPS) measured at different training steps for varying values of  $\lambda$ . SPS shows minor fluctuations but no consistent downward trend, suggesting that behavioral bias is not significantly reduced during training.

representations and output behavior. In particular, future work should explore whether bias arises from specific layers, attention patterns, or compositional effects across the model, rather than from isolated embedding directions. Developing methods to trace and intervene on these causal pathways remains a key challenge for building fair and reliable language models.

Therefore addressing this gap will be critical for developing reliable and fair language models that behave responsibly in real-world applications.

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